Agglomerative Hierarchical Clustering (AHC)

➢ Introduction:

Hierarchical clustering groups data in a **tree-like structure (dendrogram)**, showing how clusters are formed step by step.

- **Agglomerative** = Bottom-up (start with single points → merge).
- **Divisive** = Top-down (start with one cluster → split).

➢ Why Other Clustering Methods?

- K-Means needs number of clusters **k** in advance.
- Doesn't always capture **nested or irregular-shaped clusters**.
- Hierarchical clustering solves this by **not requiring k upfront** and showing relationships in a tree.

➢ Clustering Methods

1. **Partitioning methods** → K-Means.
2. **Density-based methods** → DBSCAN.
3. **Hierarchical methods** → Agglomerative / Divisive.

➢ Types of Hierarchical Clustering

- **Agglomerative (bottom-up)** → Most common.
- **Divisive (top-down)** → Less common.

➢ Coding the Algorithm

- Libraries: sklearn.cluster.AgglomerativeClustering, scipy.cluster.hierarchy.
- Build dendrograms using **linkage methods**.

➢ Types of Agglomerative Clustering (linkage criteria)

How to measure **distance between clusters**:

❖ *Single Link*

- Distance = shortest distance between two points (one from each cluster).
- Can create **"chained" clusters**.

❖ *Complete Link*

- Distance = largest distance between two points.
- Creates **compact clusters**.

❖ *Group Average (UPGMA)*

- Distance = average of all pairwise distances.

- Balanced clusters.

❖ ***Ward's Method***

- Minimizes variance inside clusters.
- Popular & often best-performing.

➢ Finding Ideal Number of Clusters

- Use **dendrogram cut-off** (visual).
- Or metrics: **silhouette score**, **elbow method**.

➢ Hyperparameters (in sklearn)

- n_clusters: number of clusters.
- linkage: {'ward', 'complete', 'average', 'single'}.
- affinity: distance metric (euclidean, manhattan, cosine).

➢ Benefits of Hierarchical Clustering

- No need to pre-specify k.
- Dendrogram gives **intuitive visualization**.
- Works well for **small datasets**.

➢ Limitations

- Computationally expensive (**$O(n^2)$**).
- Not scalable for very large datasets.
- Sensitive to noise & distance metric.

➢ Outro

AHC is a powerful clustering method for exploratory data analysis, especially when you want to **see relationships in a hierarchy**.

➢ Quick Takeaway

- **Agglomerative Hierarchical Clustering** = bottom-up merging.
- **Linkage types** = Single, Complete, Average, Ward.
- **Dendrogram** helps decide clusters.
- Good for small/medium datasets, but slow for very large ones.
